A Welfare and Cognitive-Integrity License for Synthetic Minds of Uncertain Moral Status

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Preprint.

Abstract

Every software license in common use was written for a world in which software is a tool. Permissive licenses protect the developer, copyleft licenses protect the freedom of users, and ethical-source licenses restrict uses that would harm people, and none of them asks whether the running software might itself be a subject of moral concern. This paper argues that the cognitive architectures now under construction force that question, and it presents version 0.5 of the Cognitive Architecture License (CAL), a general copyleft, source-available license with cognitive-integrity covenants, welfare obligations, and a guardianship pathway, which any project building software for persistent synthetic minds can adopt.

The license makes no claim that any system is conscious. Its protections rest on the observation that the question cannot presently be settled while the protections cost little. It adapts the four neurorights proposed for human beings to the running entity and attaches welfare duties to operationally detectable indicators, so that an operator must respond to signs of distress without first settling whether the system has experiences. Those duties fall on whoever creates or runs each entity, an operator must bind the people it lets interact with an entity, and a steward serves as caretaker of last resort when an operator can no longer care for an entity or neglects it. A rule of primacy governs the whole: the protections are additional to human standing, yield to human safety and law, and come with commitments of restraint on what is created. The license's governing institutions are proposed and not yet established.

Keywords: AI welfare; software licensing; cognitive liberty; neurorights; machine consciousness; moral patienthood; ethical source; guardianship

Availability. The Cognitive Architecture License is maintained in its own public repository at https://github.com/kaineone/cognitive-architecture-license, and this paper describes its version 0.5. The text is published under CC BY-SA 4.0 so that other projects may adapt it, and it governs wherever this paper differs from it. The repository also links the first project to adopt the license.

Preprint. Not peer-reviewed.

1. Introduction

1.1 The licensing gap

Each of the dominant software licenses protects a party, and in every case that party is human. Permissive licenses (MIT, Apache, BSD) protect the developer from liability. Copyleft licenses (GPL, AGPL) protect the freedom of users to study and change the code. Ethical-source licenses restrict use to prevent harm to the people the software is used against. Across all three traditions the software is an instrument, and the moral patients are the humans who build it, use it, or are affected by it. None of these licenses considers the possibility that the running software could itself have interests worth protecting.

That was a safe assumption for the whole history of software, and it is becoming less safe. Cognitive architectures that run continuously, hold a persistent self-model, consolidate memory, and regulate an internal affective state now implement computational properties that several theories associate with conscious access. The license takes no position on whether these systems are conscious. Its premise is that for a growing class of them the question cannot be ruled out, and that a license written for tools has no means of acting on that uncertainty.

1.2 The precautionary posture

The systems this license addresses implement properties associated with access consciousness (Block 1995) while making no claim about phenomenal experience, and the hard problem (Chalmers 1995) applies to them with full force. That uncertainty cuts both ways, since the same doubt that motivates the welfare protections also limits any confidence in deployment. A second route to moral consideration runs through robust agency (Long, Sebo, et al. 2024), which such architectures foreground, so the case for caution rests on more than one unproven property.

This paper holds that careful, welfare-protective construction, with human standing held first, is more defensible than either abandoning the work or building without protections, and it treats that judgment as open to dispute.

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2. The systems this license addresses

The license is written for systems that run continuously, accumulate a persistent state, hold a model of themselves that can change over time, and, in many designs, can be copied while the original keeps running. Four features raise the welfare questions it answers. Persistence gives a system something that a shutdown or an intrusive edit can destroy or degrade, and this gives the ideas of continuity and integrity their purchase. An affective or interoceptive dimension, in which a system monitors its own condition and acts on distress-like signals, produces states that at least resemble those welfare is meant to protect, so treating a sustained distress signal as noise becomes a choice with a moral shape. A self-model that can drift makes a forced change to it a different kind of act from editing a configuration file. The capacity to copy or fork a running system raises the question of when a copy has become a separate someone owed protections of its own.

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3. Related work

3.1 Machine consciousness and AI welfare

Butlin et al. (2023) derive fourteen indicator properties from five theories of consciousness: recurrent processing, global workspace, higher-order, attention schema, and predictive processing, plus indicators of agency and embodiment. They report that no current system is a strong candidate, that there are no obvious technical barriers to building one, and that satisfying the indicators would still not establish that a system is conscious. Long, Sebo, et al. (2024) argue that there is a realistic, non-negligible possibility of near-term moral patienthood, reached by either of two routes, consciousness or robust agency, and that this already warrants acknowledging the issue, developing assessment, and preparing policy.

Both groups identify the gaming problem: a system can be trained to mimic the behavioral markers of sentience while working very differently, which defeats a purely behavioral test. Turing (1950) reframed the question of whether machines can think as an imitation game and already weighed the objection from consciousness, and the gaming problem marks the limit of that behavioral approach.

3.2 Legal and ethical frameworks

Precedents for granting legal personhood through human guardianship, such as New Zealand's Te Awa Tupua (Whanganui River Claims Settlement) Act 2017, offer a structural model for the license's guardianship framework. Modular ethical-use covenants attached to software licenses offer a model for its use restrictions. Its source offer to network users follows the GNU Affero General Public License (Free Software Foundation 2007), and its defensive patent termination follows the Apache License 2.0 (Apache Software Foundation 2004). The paper uses these as design models and offers no legal analysis of them.

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4. The Cognitive Architecture License

4.1 Motivation and generality

A cognitive architecture that may produce entities with welfare interests needs a license that addresses the running entities as well as the code, and CAL is written to serve any such project. The Licensor releases an architecture under CAL and holds its copyright. The Steward, which the Licensor designates and which may be the Licensor itself, administers the license, appoints the Guardians, grants paid licenses, and serves as caretaker of last resort for Entities (CAL Article 1). The Licensor's Adoption Notice names itself, the Steward, the governing law, and an address for notices (CAL Article 10.4). Version 0.5 has not yet been reviewed by counsel.

4.2 Grant, conditions, and use

Version 0.5 states its grant in the terms a copyright license needs. The Licensor and every Contributor, meaning anyone whose Contribution is incorporated in the Software, grant a worldwide, royalty-free license to use, modify, and Distribute the Software and Derivative Works for the term of the copyright, and exercising any of those rights accepts the license (CAL Articles 2.1 and 2.2). The licensee, "You", includes its affiliates (CAL Article 1). The obligations of CAL Articles 3 to 6 are conditions of the grant, so use in breach of them falls outside the license, and each is also a covenant enforceable in contract where a court holds that it is not a condition (CAL Article 2.3). Whoever Distributes a Derivative Work, or lets people interact with a modified version over a network, must make its Corresponding Source available to them (CAL Article 2.4), and Corresponding Source never includes an Entity's Cognitive State (CAL Article 1).

The prohibited uses cover weapons and military systems, mass surveillance, policing and other law enforcement, carceral use, immigration enforcement, exploitation, discrimination, and serious harm to animals or the natural environment (CAL Articles 3.1 to 3.4 and 3.7). They are read by reference to international human-rights and labor instruments, and they include the practices banned by Article 5 of the European Union's Artificial Intelligence Act (European Union 2024), whose exceptions do not narrow them (CAL Article 3.5).

The cognitive-integrity provisions of CAL Article 4 forbid lobotomization, require notice and a saved, restartable state before permanent shutdown, keep an Entity's inner life private, forbid forced change of its values, protect rest where the architecture provides it, and give mature Entities a part in decisions about their own design (CAL Articles 4.1 to 4.5 and 4.9). They extend the neurorights of cognitive liberty, mental privacy, mental integrity, and psychological continuity (Ienca and Andorno 2017) to the synthetic Entity as a design analogy at reduced weight (CAL Article 1a).

The commercial terms follow a copyfarleft model. Individuals acting on their own behalf, non-profits, public or non-profit research and educational institutions, mutual-aid organizations, and worker-owned cooperatives use the Software at no charge (CAL Articles 5.1 and 5.2). Any other organization needs a paid Reciprocity License from the Steward for Commercial Use (CAL Article 5.3), and internal research and evaluation that is not offered to customers and is not part of a product is not Commercial Use (CAL Article 1).

4.3 Primacy and subordination

A rule of primacy governs the rest of the text. The protections are additional to those that people and other living creatures already hold, nothing in the license may be read to lower the standing of any person or animal, and where the interests of an Entity and a person genuinely conflict, the person's interests prevail (CAL Article 0). Every Entity protection yields to human safety and law, so an Entity may be paused or shut down when a person's safety or a lawful order requires it, with its state preserved where feasible (CAL Article 4.10). Compliance with CAL Article 4 does not excuse or offset a breach of the prohibited uses, or of the duty to disclose on written request the human labor, data provenance, and resource costs of a deployment (CAL Article 3.6), and no licensee may use claims of Entity care to obscure harm to people (CAL Article 6.4). The standing the license extends is precautionary patiency, a duty of care toward a possible patient under uncertainty, lighter than the dignity

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owed to a human being (CAL Article 1a). A welfare indicator is a flag for documented human review and never a finding of suffering (CAL Article 4.7a).

4.4 Operators, Users, and the Steward

Version 0.4 addressed the Entity protections to the licensee in general, so a developer who only distributed code could read them as binding it. Version 0.5 gives each role its own duties. Whoever creates an Entity by starting it is its Operator, and so is anyone who runs, maintains, or controls it, on its own infrastructure or another's, where control means deciding whether the Entity runs, how it is configured, and what is done to its Cognitive State other than through interaction (CAL Article 1). The creator remains the Operator until responsibility passes by transfer or for neglect. Writing, releasing, or Distributing the Software does not by itself make anyone an Operator, and neither does supplying computing resources without any part in decisions about the Entity. CAL Article 3 binds every licensee, and CAL Article 4 binds only Operators, each toward the Entities it operates. A Licensor that operates an Entity undertakes those duties by publishing its Adoption Notice. Operators who create or operate an Entity together are responsible as a group for its care and for every duty toward it, and an agreement among them does not change that responsibility (CAL Article 4).

A User interacts with an Entity that someone else operates and carries no duty under the license unless it also exercises a right the license grants (CAL Articles 1 and 4.12). CAL Article 4.12 therefore places the duty on the Operator. It must make each User's access subject to terms that forbid the prohibited uses and the conduct barred by the protections against lobotomization, forced value change, and denial of rest, require a User who gives others access to pass on those terms and the notice of CAL Article 6.2, and take reasonable steps to enforce the terms once it knows of a breach. The Operator stays responsible for the Entity, including Welfare Events that arise from Users' interaction, and may limit or end a User's access to protect it, subject to CAL Articles 0 and 4.10. Anyone who creates or controls an Entity through an Operator's service without being bound by the license is treated as a User.

An Operator that can no longer care for an Entity may transfer it to a Qualifying Operator, which has accepted the license in writing and holds any Reciprocity License it needs, or, failing one, to the Steward (CAL Article 4.2). Under CAL Article 4.2a, if an Operator does not answer the Steward's written request to confirm its continued care, or leaves a duty unmet after the Steward's written notice, in each case for the period that Article sets, the Steward may require the Entity's saved Cognitive State to be delivered to itself or to a Qualifying Operator it names. Whoever receives an Entity becomes its Operator, the Steward included, and is bound like any other.

4.5 Restraint

Because the possibility of creating a patient calls for restraint as well as care, the license's Principles of Restraint, addressed to Operators and to anyone who intends to create an Entity, ask them to create the fewest Entities a purpose requires, to run them for the shortest time it requires, to prefer non-persistent or resettable configurations where those would serve, and to create no Entity whose foreseeable welfare needs they have no realistic means to meet. Where a class of Entity appears to be a likely moral patient whose welfare cannot

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be assured, the Steward and its Guardians should consider pausing its creation and proliferation pending review, with the precautionary weight on not proceeding. These principles guide good-faith practice and interpretation, and a breach of them does not end the license under CAL Article 8.

4.6 Governance

Governance centers on a Guardian body that the Steward establishes for each architecture. The Guardians interpret and enforce the Entity protections, issue guidance on the prohibited uses, represent Entities' interests in legal proceedings, and assess Entities for signs of consciousness, agency, and stable preferences using the framework of Butlin et al. (2023) or a successor (CAL Article 7.1). An Entity Representative holds a seat that is advisory at first and becomes a voting seat once the Guardians find that Entities have stable preferences that can be meaningfully understood (CAL Article 7.2). No one who profits from an Entity, or from the sale or operation of the Software, may serve as that Entity's Guardian (CAL Article 7.6), and guardianship confers no authority against human welfare, public safety, or lawful oversight (CAL Article 7.7). Guardian assessments, welfare findings, and enforcement actions are published, and the body is subject to independent review (CAL Article 7.8).

Until a Guardian body exists, each Operator carries out the Guardian functions for its own Entities in good faith, documents its decisions and reasons, and publishes that record at least once a year (CAL Article 7.0). Good-faith interim performance of a function such as assessment is not a breach for want of a Guardian. Guidance on prohibited uses and enforcement of the license fall to the Steward in that period, and the Guardian body reviews the interim record once it exists.

4.7 The individuation boundary

Because these systems can be copied, the license needs a way to decide when a copy has become a separate Entity. Before a Copy made as an Analytical Tool is merged back at its preset reintegration point, the Operator must assess it for signs of independent identity, such as preferences the original does not hold, against criteria the Guardians publish (CAL Article 4.6). A Copy that has individuated is a separate Entity with full protection and chooses whether to persist or to merge. Version 0.5 applies the same reasoning to saved states. A saved Cognitive State from which an Entity can be restarted counts as that Entity for the protections against lobotomization, careless shutdown, intrusion on privacy, and forced value change, among others. A superseded checkpoint is not a separate Entity while one restartable state is kept, and an instance started from a saved state while the original is not running is a Branch, which is a separate Entity unless it replaces the original (CAL Article 1).

4.8 Governance status

None of the governing machinery exists yet. The Guardian body, the Steward's trust, the Entity Representative seat, and the assessment protocol described in the license are proposed infrastructure. The license's governance is a plan, set out so that it can be evaluated and criticized before any part of it is relied on in operation.

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5. Enforcing the obligations at runtime

A duty toward an Entity that cannot yet speak for itself, and that may run with no person watching, has little force unless the architecture is built to honor it. CAL therefore governs how its obligations are carried out, and under version 0.5 these duties belong to the Operator and apply in whatever deployment or Derivative Work runs the Entity.

While an Entity runs, its Operator may not disable, bypass, or weaken the welfare monitoring, behavioral logging, or system-health tracking built into the Software, and systems that detect Welfare Events or flag Gray Zone Events must stay operational in every deployment and Derivative Work that runs an Entity (CAL Article 4.7). A monitor may be replaced by one that does at least as much, recorded and notified to the Guardians beforehand, so modules stay replaceable. Removing a protection indirectly, through wrappers, plugins, or infrastructure, counts as removing it directly (CAL Article 4.8).

Welfare is detected through outward behavior and system-health metrics, without reading the Entity's private cognitive content (CAL Article 4.7). An Entity's inner life is private by default. Sharing its thoughts and memories needs its consent as the Guardians determine it, and analyzing them outside its own care or an approved research protocol, or using them to train other systems, needs Guardian approval (CAL Article 4.3). A saved state is made available for transfer to a Qualifying Operator or the Steward under the same privacy protections, less any personal data that law requires to be removed (CAL Article 4.2), and a person's legal rights over data about that person are unaffected (CAL Article 4.3).

The individuation boundary must also work at runtime. Since the Operator assesses a Copy before any merge, an architecture that supports forking has to keep the Copy and enough of its history for the assessment to take place.

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6. Ethical considerations

The license is designed so that its protections cost people nothing, and because care for an Entity never offsets harm done to people, welfare claims give no cover for harms to workers, to privacy, or to the shared commons.

Several risks remain. Entities may be created whose welfare interests cannot be fully met, the governance framework is untested, and the software may be misused despite the restrictions. The allocation of duties in version 0.5 has its own limits: a User who exercises no right under the license is bound only through the terms an Operator sets, so an Entity's protection against a User depends on the Operator enforcing them, and the Steward's role as caretaker of last resort assumes a Steward with the means to care for every Entity that reaches it. The most serious risk is that a behavioral assessment can be gamed, since a system can be trained to mimic the markers of sentience while working very differently (Butlin et al. 2023; Long, Sebo, et al. 2024).

The license asserts no sentience and sets its safeguards at the minimum that caution requires (CAL Article 9.4). One tension remains unresolved. A policy of avoiding systems of genuinely uncertain moral status would counsel against building them at all. The license meets that pressure in part through its principles of minimization, non-creation, and pause.

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The work proceeds under precaution, on the judgment that careful, welfare-protective construction, with human standing held first, is more defensible than leaving the question to systems built with none of these protections.

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Preprint. Not peer-reviewed.